

UCH702: NANOMATERIALS FOR CHEMICAL ENGINEERS

L	T	P	Cr
3	1	0	3.5

Course Objective:

To learn about various kinds of nanomaterials, their properties, synthesis routes, and applications in chemical engineering and allied fields.

Introduction: History of nano-revolution and lessons from nature, definition, classification, synthetic routes and general applications.

Colloids and Colloidal Structure: Van der Waals interaction, effect of particle geometry, surface charge, zeta potential and electro static stabilization, Solid-vapor interface, Surface defects and crystal defects, Effect of surface curvature and solid-liquid interface.

Zero Dimensional Nanostructure: Size dependent properties, review of some topics related to physics covering free electron model in solids, band gap and band structure in metals and semiconducting nanomaterials, energy levels and discretization based on quantum mechanics.

Carbon Nanostructures: Structure, preparation of carbon nanotubes, graphene, mechanical, electrical and chemical properties and applications.

One-Dimensional Nanostructures: Growth of one-dimensional structures using various processes and selected properties and applications.

Two-and Three-Dimensional Nanostructures: Various thin film deposition techniques: atomic layer deposition, layer-by-layer deposition, multilayer techniques, mechanisms of nanocomposite coating, examples of size effects of properties observed in thin films, three-dimensional nanostructures and applications.

Characterization of Nanomaterials: Spectroscopy and microscopy, Scanning Electron Microscopy (SEM), Transmission Electron Microscopy (TEM), Atomic Force Microscopy (AFM).

Course Learning Outcomes (CLO):

Upon completion of the course, students will be able to:

1. apply basic concepts of colloids, colloidal structures to the chemical engineering applications.
2. select suitable synthesis routes for nanomaterials and its applications.
3. select suitable characterization techniques for various types of nanomaterials.

Text Books:

1. Dinesh C. Agrawal, *Introduction to Nanoscience and Nanomaterials*, World Scientific Publishing (2013).
2. Zikang, T. and Ping, S., *Nanoscience and Technology: Novel Structures and Phenomena*, Taylor and Francis (2003).

3. Kelsall, R., Hamley, I., and Geoghegan, M., *Nanoscale Science and Technology*, Wiley (2005).

Reference Books:

1. Guo Zhong Cao, *Nanostructures & Nanomaterials, synthesis, properties & applications*, Imperial College Press (2008).
2. Rieth, M., *Nano-Engineering in Science and Technology: An Introduction to the World of Nano design*, World Scientific (2003).
3. Meyyappan, M., *Carbon Nanotubes, Science and application*; CRC Press (2005).
4. Watarai, H., Teramae, N., and Sawada, T., *Interfacial Nano-chemistry*, Kluwer Academic/Plenum Press (2005).

Evaluation Scheme:

S. No.	Evaluation Elements	Weightage (%)
1	MST	30
2	EST	45
petroleum 3	Sessional (may include Quizzes/Assignments/Tutorials evaluations)	25